EXPERIMENT NO. 06

Visualize Datasets Using Bar Chart, Histogram, Pie Chart, and Boxplot

PART B

TO BE COMPLETED BY STUDENTS

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Date of Experiment:	Date of Submission
Grade:	

B.1 Program Code

```
#------------------------------------------------
# EXPERIMENT NO. 06
# VISUALIZATION USING BAR CHART, HISTOGRAM,
# PIE CHART AND BOXPLOT
#------------------------------------------------

# Install ggplot2 if required
# install.packages("ggplot2")

# Load library
library(ggplot2)

#------------------------------------------------
# CREATE AGRICULTURE DATASET
#------------------------------------------------

agri_data <- data.frame(
  Crop = c(
    "Rice",
    "Wheat",
    "Maize",
    "Cotton",
    "Sugarcane",
    "Soybean",
    "Groundnut",
```

```
    "Pulses",
    "Jowar",
    "Bajra"
  ),

  Production = c(
    500,
    400,
    350,
    300,
    600,
    250,
    280,
    220,
    180,
    160
  ),

  Area = c(
    100,
    80,
    70,
    90,
    120,
    75,
    70,
    60,
    50,
    45
  ),

  Yield = c(
    5.0,
    5.0,
    5.0,
    3.3,
    5.0,
    3.3,
    4.0,
    3.7,
    3.6,
    3.6
  )
)

# Display dataset
print(agri_data)

# Display first observations
head(agri_data)

# Display summary
summary(agri_data)

#-------------------------------------------------
# 1. BAR CHART
# Crop-wise Production
#-------------------------------------------------

ggplot(
  agri_data,
```

```
  aes(x = Crop, y = Production)
) +
  geom_col() +
  labs(
    title = "Crop-wise Agricultural Production",
    x = "Crop",
    y = "Production"
  ) +
  theme_minimal()

#-------------------------------------------------
# 2. HISTOGRAM
# Distribution of Production
#-------------------------------------------------

ggplot(
  agri_data,
  aes(x = Production)
) +
  geom_histogram(
    bins = 5
  ) +
  labs(
    title = "Distribution of Agricultural Production",
    x = "Production",
    y = "Frequency"
  ) +
  theme_minimal()

#-------------------------------------------------
# 3. PIE CHART
# Crop-wise Share of Production
#-------------------------------------------------

ggplot(
  agri_data,
  aes(
    x = "",
    y = Production,
    fill = Crop
  )
) +
  geom_col() +
  coord_polar("y") +
  labs(
    title = "Crop-wise Share of Total Production"
  ) +
  theme_void()

#-------------------------------------------------
# 4. BOXPLOT
# Distribution of Production
#-------------------------------------------------

ggplot(
  agri_data,
  aes(
    x = "Agricultural Crops",
    y = Production
  )
) +
```

```
  geom_boxplot() +
  labs(
    title = "Boxplot of Agricultural Production",
    x = "",
    y = "Production"
  ) +
  theme_minimal()

#-------------------------------------------------
# 5. BOXPLOT
# Yield Distribution
#-------------------------------------------------

ggplot(
  agri_data,
  aes(
    x = "Agricultural Crops",
    y = Yield
  )
) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Crop Yield",
    x = "",
    y = "Yield"
  ) +
  theme_minimal()
```

B.2 Expected Output

The program generates the following visualizations.

Output 1: Bar Chart

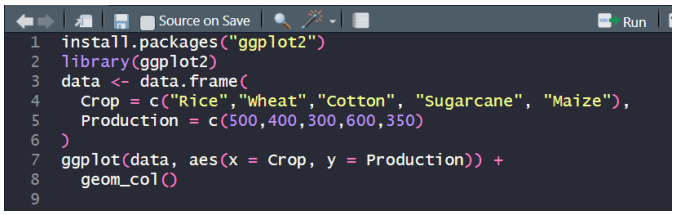
Title: Crop-wise Agricultural Production

The bar chart displays production values for different crops.

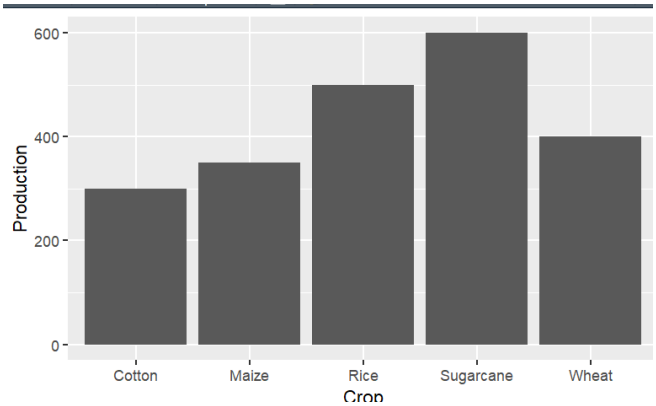
Interpretation

The graph makes it easy to compare production among different crops. In the given sample data, sugarcane has the highest production value, while bajra has the lowest.

Input:-



Output:-



Output 2: Histogram

Title: Distribution of Agricultural Production

The histogram represents the frequency distribution of production values.

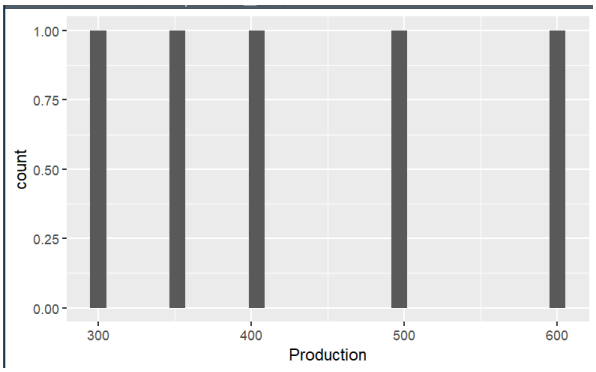
Interpretation

The histogram helps understand how the production observations are distributed across different ranges.

Input:-

```
ggplot(data, aes(x = Production)) +
  geom_histogram()
```

Output:-



Output 3: Pie Chart

Title: Crop-wise Share of Total Production

The pie chart represents the contribution of each crop to the total production.

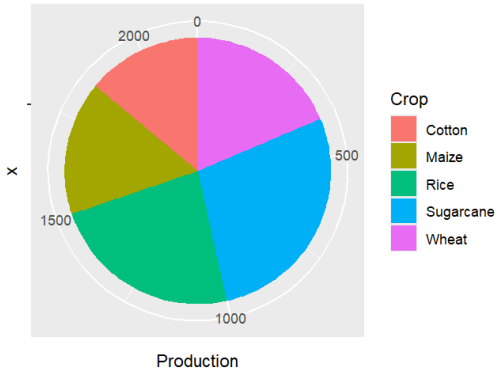
Interpretation

Each slice represents the proportion of total production contributed by a particular crop.

Input:-

```
ggplot(data, aes(x = "", y = Production, fill = Crop)) +
  geom_col() +
  coord_polar("y")
```

Output:-



Output 4: Boxplot

Title: Boxplot of Agricultural Production

The boxplot summarizes the distribution of production values.

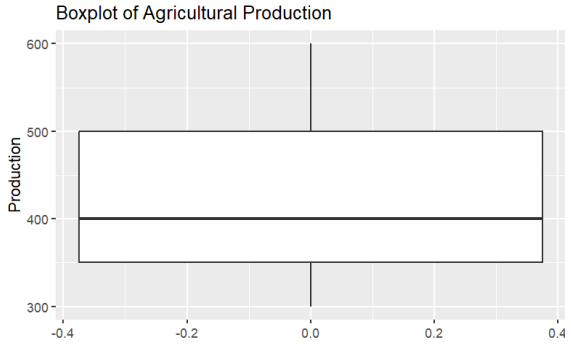
Interpretation

The boxplot displays the median, quartiles, spread, and possible outliers in the production data.

Input:-

```
ggplot(data, aes(y = Production)) +
  geom_boxplot() +
  labs(title = "Boxplot of Agricultural Production",
       y = "Production")
```

Output:-



Output 5: Yield Boxplot

Title: Boxplot of Crop Yield

This graph represents the distribution of crop yield.

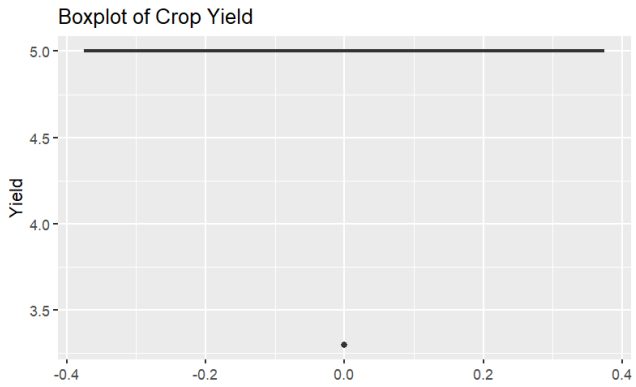
Interpretation

The boxplot can be used to study the spread and variation in yield values.

Input:-

```
data <- data.frame(
  Crop = c("Rice", "Wheat", "Cotton", "Sugarcane", "Maize"),
  Production = c(500, 400, 300, 600, 350),
  Yield = c(5.0, 5.0, 3.3, 5.0, 5.0)
)
ggplot(data, aes(y = Yield)) +
  geom_boxplot() +
  labs(title = "Boxplot of Crop Yield",
       y = "Yield")
```

Output:-



B.3 Observation

Sr. No.	Visualization	Observation
1	Bar Chart	Crop-wise production was compared successfully.
2	Histogram	Distribution of agricultural production was visualized.
3	Pie Chart	Contribution of each crop to total production was represented.
4	Boxplot	Distribution and spread of production were analyzed.
5	Yield Boxplot	Variation in crop yield was represented.

B.4 Observation Table

Graph	Variable	Purpose	Result
Bar Chart	Crop & Production	Compare crop production	Successfully visualized
Histogram	Production	Study frequency distribution	Successfully visualized
Pie Chart	Crop & Production	Show production share	Successfully visualized
Boxplot	Production	Study spread and outliers	Successfully visualized
Boxplot	Yield	Study yield variation	Successfully visualized

B.5 Result

The agriculture dataset was successfully visualized in R using **bar chart, histogram, pie chart, and boxplot**.

The bar chart was used to compare crop-wise production, the histogram was used to understand the distribution of production values, the pie chart represented the contribution of each crop to total production, and the boxplot was used to study the spread and variation of the dataset.

Thus, the experiment demonstrated the practical application of graphical data visualization techniques for statistical data analysis.

B.6 Applications

The visualization techniques used in this experiment can be applied to:

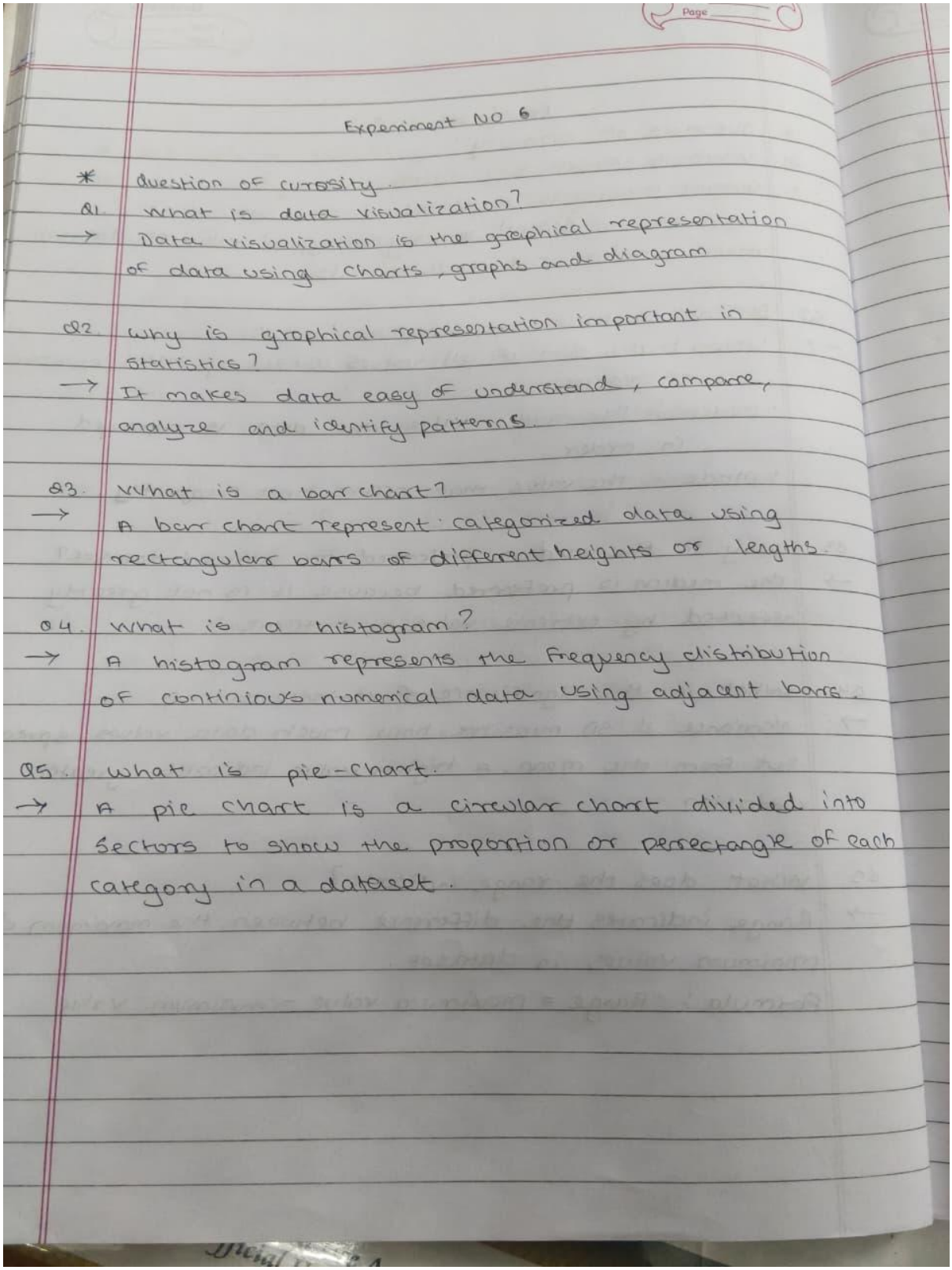
1. Agricultural production analysis.
2. Crop yield comparison.
3. Smart city data analysis.
4. Air quality analysis.
5. Population analysis.
6. Traffic analysis.
7. Healthcare data analysis.
8. Business and sales analysis.
9. Environmental data analysis.
10. Financial data analysis.
11. Exploratory data analysis.
12. Statistical reporting.
13. AI and machine learning data analysis.
14. Decision-making and planning.

The syllabus identifies data visualization and interpretation of analytical results as an important laboratory outcome.

B.7 Precautions

1. Ensure that `ggplot2` is installed and loaded correctly.
2. Check the dataset before visualization.
3. Ensure numerical variables are stored in numeric format.
4. Select an appropriate graph according to the type of data.
5. Use histograms for numerical distributions.
6. Use bar charts for categorical comparisons.
7. Use pie charts only when categories represent meaningful parts of a whole.
8. Add meaningful titles and axis labels.
9. Select a suitable number of histogram bins.
10. Check boxplots for possible outliers.
11. Verify the generated graph before recording the result.
12. Save the R program and screenshots of all outputs.

B.8 Questions of Curiosity



Experiment NO 6

* Question of curosity.

Q1 What is data visualization?

→ Data visualization is the graphical representation of data using charts, graphs and diagram

Q2 Why is graphical representation important in statistics?

→ It makes data easy of understand, compare, analyze and identify patterns

Q3 What is a bar chart?

→ A bar chart represent categorized data using rectangular bars of different heights or lengths.

Q4 What is a histogram?

→ A histogram represents the freqyency distribution of continious numerical data using adjacent bars.

Q5 What is pie-chart.

→ A pie chart is a circular chart divided into sectors to show the proportion or perrectangle of each category in a dataset.

B.9 Viva Voce Questions

1. What is R?
2. What is RStudio?
3. What is ggplot2?
4. How do you install ggplot2?
5. How do you load ggplot2?
6. What is the purpose of `ggplot()`?
7. What is `aes()`?
8. What is a geom?
9. Which function is used to create a bar chart?
10. Which function is used to create a histogram?
11. Which function is used to create a boxplot?
12. How is a pie chart created in ggplot2?
13. What is `coord_polar()`?
14. What is the difference between `geom_bar()` and `geom_col()`?
15. What does a histogram show?
16. What does a boxplot show?
17. What is Q1?
18. What is Q3?
19. What is IQR?
20. What is an outlier?
21. Why are graphs useful in statistical analysis?
22. Which graph is suitable for comparing categories?
23. Which graph is suitable for showing data distribution?
24. Which graph can be used to show parts of a whole?
25. Which graph is useful for identifying outliers?
26. Give two applications of data visualization in agriculture.
27. Give two applications of data visualization in smart cities.
28. Why should graphs have proper labels?
29. What is exploratory data analysis?
30. How can visualization help in decision-making?